

Triple-Higgs-boson probes of the Higgs potential and top-Higgs-boson interactions

John M. Campbell¹, Giuseppe De Laurentis², R. Keith Ellis³, Michelle Gensmann^{4,5},
Ulrich Haisch^{4*}, Luca Panizzi^{6†}, Andreas Papaefstathiou^{7‡} and
Gilberto Tetlalmatzi-Xolocotzi^{8§}

¹ Fermilab, PO Box 500, Batavia IL 60510-5011, USA

² Higgs Centre for Theoretical Physics, University of Edinburgh, Edinburgh, EH9 3FD, UK

³ Institute for Particle Physics Phenomenology, Durham University, Durham, DH1 3LE, UK

⁴ Max Planck Institute for Physics, Boltzmannstr. 8, 85748 Garching, Germany

⁵ Institute for Theoretical Physics, Karlsruhe Institute of Technology, Wolfgang-Gaede-Str. 1, 76131 Karlsruhe, Germany

⁶ Dipartimento di Fisica, Università della Calabria, I-87036 Arcavacata di Rende, Cosenza, Italy

⁷ Department of Physics, Kennesaw State University, 830 Polytechnic Lane, Marietta, GA 30060, USA

⁸ Theoretische Physik 1, Center for Particle Physics Siegen (CPPS), Universität Siegen, Walter-Flex-Str. 3, 57068 Siegen, Germany

★ haisch@mpp.mpg.de

† luca.panizzi@unical.it

‡ apapaefs@kennesaw.edu

§ gtx@physik.uni-siegen.de

Preprint: FERMILAB-PUB-26-0464-T, LHCHWG-2026-xxx, MPP-2026-xxx

Abstract

We study triple-Higgs-boson production in the κ -framework as a direct probe of the Higgs potential. We derive complementary compact parametrisations of the inclusive gluon-fusion cross section: one retains the dependence on κ_3 , κ_4 , κ_t , and κ_{2t} for $\kappa_{3t} = 0$, while the other incorporates κ_{3t} for $\kappa_t = 1$ and $\kappa_{2t} = 0$. These results provide practical tools for interpreting current and future LHC measurements. We use the parametrisations to assess the sensitivity of LHC Run 2 data and HL-LHC projections to the Higgs boson self-couplings and show that differential kinematic distributions can provide additional discriminatory power between scenarios with nearly identical inclusive production rates.

Copyright attribution to authors.

This work is a submission to SciPost Physics.

License information to appear upon publication.

Publication information to appear upon publication.

Received Date

Accepted Date

Published Date

Contents

1 Introduction	2
2 Triple-Higgs-boson production in the κ-framework	3
2.1 Including the two-top–three-Higgs-boson interaction	8
3 Conclusions	12
References	13

1 Introduction

The discovery of the Higgs boson in 2012 [1, 2] completed the particle content of the Standard Model (SM) and established the mechanism of electroweak (EW) symmetry breaking. Since then, a major objective of the LHC physics programme has been to determine the properties of the Higgs boson with increasing precision. While its couplings to EW gauge bosons and third-generation fermions are already constrained at the few- to ten-percent level, its self-interactions remain largely unexplored. Measuring the Higgs **boson** self-couplings is therefore one of the most important outstanding goals of collider physics, since they directly probe the shape of the Higgs potential and the dynamics of EW symmetry breaking.

In the SM, the Higgs potential is determined by a single quartic coupling $\lambda = m_h^2/(2v^2)$. After EW symmetry breaking, it predicts unique values for the Higgs **boson** trilinear and quartic self-couplings,

$$\lambda_3^{\text{SM}} = \lambda, \quad \lambda_4^{\text{SM}} = \frac{\lambda}{4}, \quad (1)$$

once the Higgs boson mass $m_h \simeq 125$ GeV and vacuum expectation value $v \simeq 246$ GeV are fixed. Any deviation from these predictions would provide evidence for physics beyond the SM, such as extended scalar sectors, composite Higgs models, or mechanisms for EW baryogenesis. Reviews of such beyond-the-SM (BSM) scenarios can be found, for example, in Refs. [3, 4].

At the LHC, **double-Higgs-boson** production provides the primary direct probe of the Higgs **boson** trilinear self-coupling, while loop-induced effects in **single-Higgs-boson** processes yield complementary constraints [5–13]. These measurements are typically interpreted in the κ -framework [14–16], which parametrises deviations from the SM through

$$\kappa_3 = \frac{\lambda_3}{\lambda_3^{\text{SM}}}, \quad (2)$$

assuming all other couplings take their SM values. Using the full $\sqrt{s} = 13$ TeV LHC Run 2 data set, the ATLAS and CMS collaborations obtain the combined constraint

$$-0.71 < \kappa_3 < 6.1 \quad (\text{ATLAS} + \text{CMS}) \quad (3)$$

at 95% confidence level (CL) [17]. The HL-LHC, with an integrated luminosity of **3 ab⁻¹ per experiment, corresponding to a combined ATLAS–CMS exposure of 6 ab⁻¹**, is expected to improve this sensitivity to

$$0.5 < \kappa_3 < 1.7 \quad (\text{HL-LHC}). \quad (4)$$

This projection is taken from Ref. [50].

Constraining the quartic Higgs **boson** self-coupling is considerably more challenging. It is conventionally parametrised as

$$\kappa_4 = \frac{\lambda_4}{\lambda_4^{\text{SM}}}, \quad (5)$$

and **triple-Higgs-boson** production provides its only direct probe at hadron colliders. However, the tiny SM cross section of approximately 100 ab at $\sqrt{s} = 14$ TeV, calculated at next-to-next-to-leading order (NNLO) in QCD [18], renders such measurements experimentally very challenging. Current ATLAS [19] and CMS [20] searches yield

$$-230 < \kappa_4 < 240 \quad (\text{ATLAS}), \quad -190 < \kappa_4 < 190 \quad (\text{CMS}), \quad (6)$$

at 95% CL, assuming all other couplings are SM-like, in particular $\kappa_3 = 1$. A recent HL-LHC study [21] projects these bounds to improve to

$$-81 < \kappa_4 < 89 \quad (\text{HL-LHC}) \quad (7)$$

for a combined ATLAS–CMS exposure of 6 ab^{-1} , corresponding to 3 ab^{-1} per experiment. In addition to **triple-Higgs-boson** production, **double-Higgs-boson** production is also sensitive to κ_4 through two-loop EW corrections [21–24], although these constraints are generally weaker.

In this note, we present a comprehensive study of **triple-Higgs-boson** production within the κ -framework. We derive **complementary** compact parametrisations of the inclusive gluon-fusion **cross section**. The first retains the dependence on the Higgs **boson** trilinear and quartic self-couplings, the top-quark Yukawa coupling modifier κ_t , and the **two-top–two-Higgs-boson** interaction κ_{2t} for $\kappa_{3t} = 0$. The second incorporates the **two-top–three-Higgs-boson** interaction κ_{3t} for $\kappa_t = 1$ and $\kappa_{2t} = 0$. These expressions provide practical tools for interpreting current and future LHC measurements. Using them, we investigate the sensitivity of current **LHC data and HL-LHC projections** to the Higgs **boson** self-couplings and study the impact of correlated modifications of **the top–Higgs-boson interactions**. We further demonstrate that differential kinematic distributions retain significant sensitivity even in scenarios with nearly degenerate inclusive cross sections, highlighting the importance of shape information in future **triple-Higgs-boson** searches.

2 Triple-Higgs-boson production in the κ -framework

Triple-Higgs-boson production at hadron colliders has been the subject of extensive theoretical investigation over the past two decades [18, 21, 22, 25–41]. Within the SM, the gluon-fusion process $gg \rightarrow 3h$ is dominated by the one-loop topologies shown in the upper and middle rows of Figure 1. In the figure, green (blue) boxes denote insertions of the Higgs **boson** trilinear (quartic) self-coupling, while orange circles represent the top-quark Yukawa coupling. The two diagrams in the lower row illustrate representative BSM contributions involving a contact interaction between two top quarks and two Higgs bosons. In the SM, bottom-quark loops **modify** the $gg \rightarrow 3h$ production cross section **by only about 0.2%** at LHC energies [39]; **they are nevertheless retained in the numerical calculations below**.

The **relevant terms in the κ -framework Lagrangian** are

$$\mathcal{L}_\kappa \supset -\kappa_3 v \lambda h^3 - \kappa_4 \frac{\lambda}{4} h^4 - \kappa_t \frac{y_t}{\sqrt{2}} \bar{t} t h - \kappa_{2t} \frac{y_t}{\sqrt{2} v} \bar{t} t h^2 - \kappa_{3t} \frac{y_t}{2\sqrt{2} v^2} \bar{t} t h^3, \quad (8)$$

where κ_3 and κ_4 were introduced in Eqs. (2) and (5), respectively, and $y_t = \sqrt{2} m_t / v$, with m_t denoting the top-quark mass. The parameter κ_t thus encodes modifications of the top-quark Yukawa coupling, while κ_{2t} and κ_{3t} **parametrise contact interactions** between two top

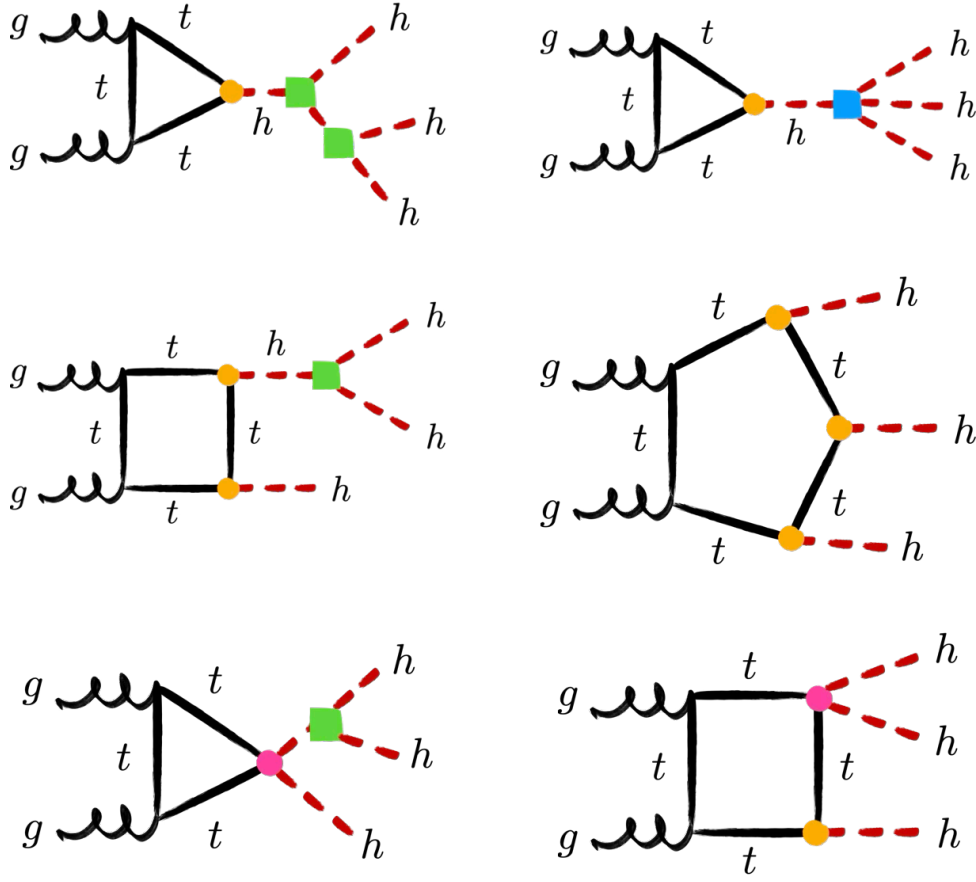


Figure 1: Representative one-loop Feynman diagrams contributing to **triple-Higgs-boson** production via gluon fusion, illustrating the triangle, box, and pentagon topologies. Green (blue) boxes denote the Higgs **boson** trilinear (quartic) self-coupling, while orange (magenta) circles indicate top-quark couplings involving one (two) Higgs bosons.

quarks and two **or three Higgs bosons, respectively**. The final term can equivalently be written as $-\kappa_{3t}m_t\bar{t}th^3/(2v^3)$ [35]. The SM is recovered from Eq. (8) for $\kappa_3 = \kappa_4 = \kappa_t = 1$ and $\kappa_{2t} = \kappa_{3t} = 0$.

To determine the dependence of the inclusive $gg \rightarrow 3h$ production cross section on κ_3 , κ_4 , κ_t , and κ_{2t} , we employ the leading-order (LO) matrix elements calculated in Ref. [39] and implemented in the **MCFM** [42–45] parton-level event generator. We use $m_h = 125$ GeV and $m_t = 173$ GeV as input parameters together with the **PDF4LHC21** parton distribution functions (PDFs) [46]. The renormalisation and factorisation scales, μ_R and μ_F , are chosen dynamically as $\mu_R = \mu_F = m_{3h}/2$, where m_{3h} denotes the invariant mass of the **three-Higgs-boson system**. At **centre-of-mass energies** $\sqrt{s} = 13$ TeV, **13.6 TeV**, and **14 TeV**, and with $\kappa_{3t} = 0$, we parametrise the LO cross section as

$$\sigma_{\text{LO}}^{\sqrt{s}} = (\sigma_{\text{LO}}^{\sqrt{s}})_{\text{SM}} \sum_{n,m} c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t}) (\Delta\kappa_3)^n (\Delta\kappa_4)^m, \quad (9)$$

where $\Delta\kappa_i = \kappa_i - 1$ for $i = 3, 4$. The corresponding LO SM cross sections are

$$\sigma_{\text{LO}}^{13\text{TeV}} = 42.1 \text{ ab}, \quad \sigma_{\text{LO}}^{13.6\text{TeV}} = 47.9 \text{ ab}, \quad \sigma_{\text{LO}}^{14\text{TeV}} = 51.0 \text{ ab}. \quad (10)$$

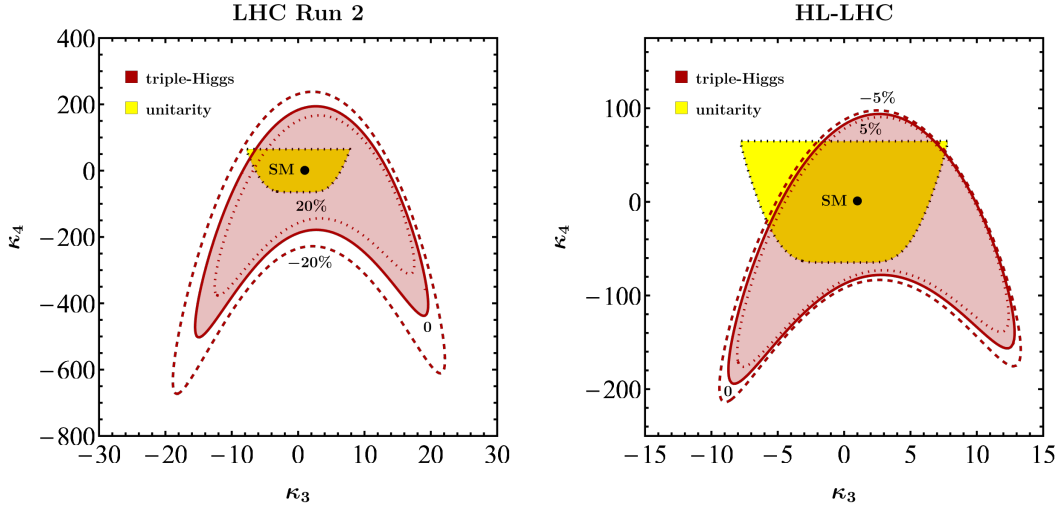


Figure 2: Constraints in the κ_3 - κ_4 plane at LHC Run 2 (left) and the HL-LHC (right). The red contours indicate the **allowed** 95% CL regions obtained from **triple-Higgs-boson** production in gluon fusion for three representative values of κ_t , as specified by the contour labels. We assume $\kappa_{2t} = \kappa_{3t} = 0$. The SM prediction is denoted by black points. The yellow regions, enclosed by black dotted contours, represent the **perturbative-unitarity** bounds derived from tree-level $hh \rightarrow hh$ scattering.

For $\sqrt{s} = 13$ TeV, the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eq. (9) are

$$c_{00}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = -6.05 \times 10^{-4} + 0.0384\kappa_t^2 - 0.350\kappa_t^3 + 1.47\kappa_t^4 - 3.04\kappa_t^5 + 2.89\kappa_t^6 \\ + \kappa_{2t} (1.61\kappa_t^2 - 7.82\kappa_t^3 + 15.6\kappa_t^4 - 14.8\kappa_t^5) \\ + \kappa_{2t}^2 (6.96 - 22.3\kappa_t + 37.0\kappa_t^2), \quad (11)$$

$$c_{01}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0829\kappa_t^2 - 0.264\kappa_t^3 + 0.0895\kappa_t^4 + \kappa_{2t} (0.673\kappa_t - 1.12\kappa_t^2), \quad (12)$$

$$c_{02}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0169\kappa_t^2, \quad (13)$$

$$c_{10}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.211\kappa_t^2 - 1.45\kappa_t^3 + 3.65\kappa_t^4 - 3.28\kappa_t^5 \\ + \kappa_{2t} (3.49\kappa_t - 13.4\kappa_t^2 + 15.6\kappa_t^3) + \kappa_{2t}^2 (13.9 - 22.3\kappa_t), \quad (14)$$

$$c_{11}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0978\kappa_t^2 - 0.264\kappa_t^3 + 0.673\kappa_{2t}\kappa_t, \quad (15)$$

$$c_{20}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.288\kappa_t^2 - 1.25\kappa_t^3 + 1.84\kappa_t^4 + \kappa_{2t} (2.81\kappa_t - 6.70\kappa_t^2) + 6.96\kappa_{2t}^2, \quad (16)$$

$$c_{21}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0489\kappa_t^2, \quad (17)$$

$$c_{30}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.159\kappa_t^2 - 0.417\kappa_t^3 + 0.938\kappa_{2t}\kappa_t, \quad (18)$$

$$c_{40}^{13\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0396\kappa_t^2. \quad (19)$$

At $\sqrt{s} = 13.6$ TeV, the corresponding coefficients read

$$c_{00}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = -9.34 \times 10^{-3} + 0.0572\kappa_t^2 - 0.446\kappa_t^3 + 1.68\kappa_t^4 - 3.17\kappa_t^5 + 2.89\kappa_t^6 \\ + \kappa_{2t} (1.59\kappa_t^2 - 7.69\kappa_t^3 + 15.3\kappa_t^4 - 14.6\kappa_t^5) \\ + \kappa_{2t}^2 (6.89 - 22.1\kappa_t + 36.8\kappa_t^2), \quad (20)$$

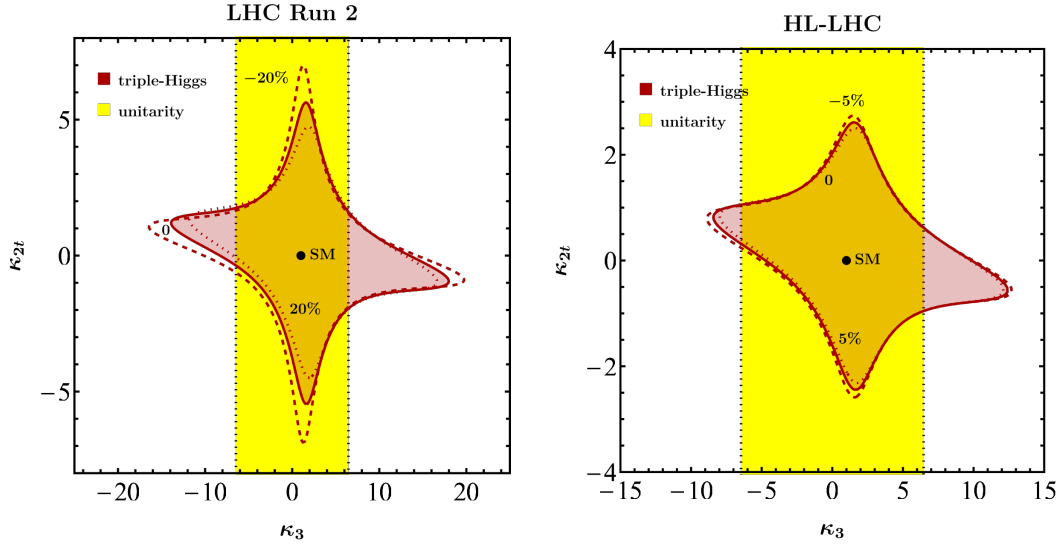


Figure 3: As in Figure 2, but in the κ_3 - κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). The parameters not shown explicitly are fixed to their SM values, i.e. $\kappa_4 = \kappa_t = 1$ and $\kappa_{3t} = 0$.

$$c_{01}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0814\kappa_t^2 - 0.260\kappa_t^3 + 0.0866\kappa_t^4 + \kappa_{2t}(0.665\kappa_t - 1.10\kappa_t^2), \quad (21)$$

$$c_{02}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0167\kappa_t^2, \quad (22)$$

$$c_{10}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.204\kappa_t^2 - 1.40\kappa_t^3 + 3.56\kappa_t^4 - 3.21\kappa_t^5 + \kappa_{2t}(3.43\kappa_t - 13.2\kappa_t^2 + 15.3\kappa_t^3) + \kappa_{2t}^2(13.8 - 22.1\kappa_t), \quad (23)$$

$$c_{11}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0961\kappa_t^2 - 0.260\kappa_t^3 + 0.665\kappa_{2t}\kappa_t, \quad (24)$$

$$c_{20}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.283\kappa_t^2 - 1.23\kappa_t^3 + 1.81\kappa_t^4 + \kappa_{2t}(2.77\kappa_t - 6.59\kappa_t^2) + 6.89\kappa_{2t}^2, \quad (25)$$

$$c_{21}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0480\kappa_t^2, \quad (26)$$

$$c_{30}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.155\kappa_t^2 - 0.408\kappa_t^3 + 0.922\kappa_{2t}\kappa_t, \quad (27)$$

$$c_{40}^{13.6\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0388\kappa_t^2. \quad (28)$$

Finally, for $\sqrt{s} = 14\text{TeV}$, we obtain

$$c_{00}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = -2.18 \times 10^{-3} + 0.0323\kappa_t^2 - 0.311\kappa_t^3 + 1.35\kappa_t^4 - 2.92\kappa_t^5 + 2.84\kappa_t^6 + \kappa_{2t}(1.61\kappa_t^2 - 7.79\kappa_t^3 + 15.5\kappa_t^4 - 14.7\kappa_t^5) + \kappa_{2t}^2(6.99 - 22.4\kappa_t + 37.4\kappa_t^2), \quad (29)$$

$$c_{01}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0822\kappa_t^2 - 0.262\kappa_t^3 + 0.0865\kappa_t^4 + \kappa_{2t}(0.672\kappa_t - 1.12\kappa_t^2), \quad (30)$$

$$c_{02}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0169\kappa_t^2, \quad (31)$$

$$c_{10}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.194\kappa_t^2 - 1.39\kappa_t^3 + 3.58\kappa_t^4 - 3.24\kappa_t^5 + \kappa_{2t}(3.46\kappa_t - 13.3\kappa_t^2 + 15.5\kappa_t^3) + \kappa_{2t}^2(14.0 - 22.4\kappa_t), \quad (32)$$

$$c_{11}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0968\kappa_t^2 - 0.262\kappa_t^3 + 0.672\kappa_{2t}\kappa_t, \quad (33)$$

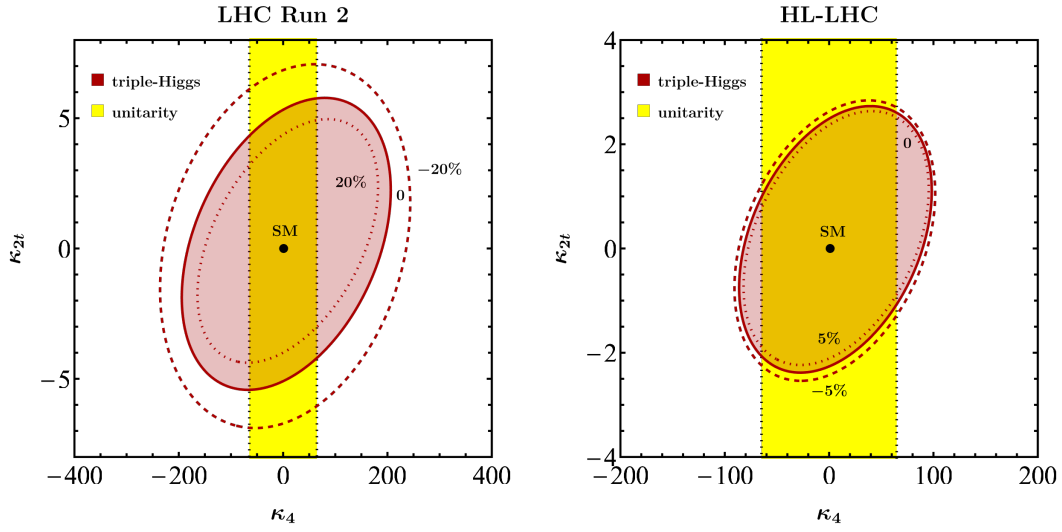


Figure 4: As in Figure 2, but in the κ_4 - κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). The remaining parameters are fixed to their SM values, i.e. $\kappa_3 = \kappa_t = 1$ and $\kappa_{3t} = 0$.

$$c_{20}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.284\kappa_t^2 - 1.23\kappa_t^3 + 1.82\kappa_t^4 + \kappa_{2t}(2.79\kappa_t - 6.65\kappa_t^2) + 6.99\kappa_{2t}^2, \quad (34)$$

$$c_{21}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0484\kappa_t^2, \quad (35)$$

$$c_{30}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.156\kappa_t^2 - 0.411\kappa_t^3 + 0.931\kappa_{2t}\kappa_t, \quad (36)$$

$$c_{40}^{14\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0391\kappa_t^2. \quad (37)$$

We note that the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ depend only mildly on the collision energy over the range considered. Likewise, the choice of PDFs is expected to have a negligible numerical impact on the parametrisations given in Eqs. (11)–(37). Higher-order QCD corrections to the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ are currently unknown, although they could, in principle, be estimated in the infinite-top-quark-mass limit. In the SM, these corrections increase the inclusive cross section by a K -factor of approximately 1.7 at NNLO in QCD [18]. Parametrisations of the triple-Higgs-boson gluon-fusion cross section at the LHC similar to Eqs. (9)–(37) have also been presented in Refs. [35, 36, 38, 39].

The two panels in Figure 2 illustrate the sensitivity to the Higgs boson self-couplings κ_3 and κ_4 using the full LHC Run 2 data set and HL-LHC projections corresponding to 3ab^{-1} per experiment. The red contours indicate the 95% CL constraints derived from triple-Higgs-boson production in gluon fusion using the parametrisations in Eqs. (11)–(19) and Eqs. (29)–(37), with the normalisation to the corresponding LO SM cross section given in Eq. (10). For LHC Run 2, the triple-Higgs-boson constraints are obtained by imposing $\mu_{3h}^{\text{LHC Run 2}} < 588$ [20], while for the HL-LHC we assume a projected upper limit of $\mu_{3h}^{\text{HL-LHC}} < 125$ [47]. Throughout this analysis, we set $\kappa_{2t} = \kappa_{3t} = 0$, and the black points indicate the SM prediction. The left panel demonstrates that modifications of the top-quark Yukawa coupling of $\Delta\kappa_t = \pm 20\%$, where $\Delta\kappa_t = \kappa_t - 1$, which remain allowed at 95% CL after LHC Run 2 [48, 49], have a significant impact on the resulting constraints. This behaviour was already pointed out in Ref. [41] and originates from the strong power-law dependence of the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eqs. (11)–(37) on κ_t . By the end of the HL-LHC, the top-quark Yukawa coupling is expected to be measured with a precision of $\Delta\kappa_t = \pm 5\%$ at 95% CL [50]. Consequently, its impact on the $gg \rightarrow 3h$ constraints in the κ_3 - κ_4 plane will be substantially reduced, as illustrated in the right panel. Finally, we note that the projected HL-LHC constraints from $gg \rightarrow 3h$

are comparable to those imposed by perturbative unitarity [34, 51, 52], shown as the yellow region enclosed by the black-dotted contour in both panels.

The panels in Figures 3 and 4 show the expected sensitivity in the κ_3 – κ_{2t} and κ_4 – κ_{2t} planes, respectively. The red contours represent the 95% CL constraints from **triple-Higgs-boson** production in gluon fusion, obtained following the procedure described above. Parameters not shown explicitly are fixed to their SM values, and the black **points** indicate the SM prediction. As in Figure 2, the left panels demonstrate that variations of the top-quark Yukawa coupling of $\Delta\kappa_t = \pm 20\%$ can have a significant impact on the resulting constraints, whereas the smaller variations $\Delta\kappa_t = \pm 5\%$ considered in the right panels lead to only minor changes. For κ_{2t} , **triple-Higgs-boson** production at LHC Run 2 yields $-5.1 < \kappa_{2t} < 5.3$, while the HL-LHC is expected to improve this range to $-2.3 < \kappa_{2t} < 2.5$ at 95% CL. These limits are obtained by fixing $\kappa_3 = \kappa_4 = \kappa_t = 1$ and $\kappa_{3t} = 0$. We note that, under the same assumptions, the 95% CL constraint on the **double-Higgs-boson** production signal strength obtained at LHC Run 2, $\mu_{2h}^{\text{LHC Run 2}} < 2.5$ [17], already implies $-0.19 < \kappa_{2t} < 0.72$. The quoted bound is obtained using the expressions given in Refs. [53, 54]. Consequently, **triple-Higgs-boson** production is not expected to provide a stronger constraint on κ_{2t} than **double-Higgs-boson** production in a single-parameter analysis. It can, however, provide complementary information in multi-parameter fits, in particular when these involve κ_4 , which can only be weakly constrained by **double-Higgs-boson** production [21–24].

2.1 Including the two-top–three-Higgs-boson interaction

We next extend the inclusive-rate parametrisation to the contact interaction between two top quarks and three Higgs bosons in Eq. (8). We generate loop-induced LO parton-level samples with MadGraph5_aMC@NLO [55], using the same Higgs boson mass m_h , top-quark mass m_t , PDFs, and scale choice as above. In this scan, $\kappa_t = 1$ and $\kappa_{2t} = 0$ are fixed.

Following Eq. (9), we write the extended cross section as

$$\sigma_{\text{LO}}^{\sqrt{s}} = (\sigma_{\text{LO}}^{\sqrt{s}})_{\text{SM}} \sum_{n,m} c_{nm}^{\sqrt{s}}(\kappa_{3t}) (\Delta\kappa_3)^n (\Delta\kappa_4)^m. \quad (38)$$

This is an exact regrouping of the LO coupling basis at the fixed values $\kappa_t = 1$ and $\kappa_{2t} = 0$. Since the amplitude is linear in κ_{3t} , c_{00} is at most quadratic in κ_{3t} , while c_{01} , c_{10} , and c_{20} are at most linear; the remaining nonzero coefficient functions are constant. The fitted SM cross sections are

$$(\sigma_{\text{LO}}^{13\text{TeV}})_{\text{SM}} = 42.12 \text{ ab}, \quad (\sigma_{\text{LO}}^{13.6\text{TeV}})_{\text{SM}} = 47.40 \text{ ab}, \quad (\sigma_{\text{LO}}^{14\text{TeV}})_{\text{SM}} = 51.08 \text{ ab}. \quad (39)$$

They agree to within 1.1% with the independent values in Eq. (10).

The normalisation fixes $c_{00}^{\sqrt{s}}(0) = 1$. The fitted coefficient functions are listed below. Their additional decimal places, compared with the first parametrisation, are retained to reproduce the numerical accuracy of the fit.

For $\sqrt{s} = 13 \text{ TeV}$, the non-vanishing coefficient functions are

$$c_{00}^{13\text{TeV}}(\kappa_{3t}) = 1 - 2.83183 \kappa_{3t} + 15.81829 \kappa_{3t}^2, \quad (40)$$

$$c_{01}^{13\text{TeV}}(\kappa_{3t}) = -0.0914560 + 0.793591 \kappa_{3t}, \quad (41)$$

$$c_{02}^{13\text{TeV}}(\kappa_{3t}) = 0.0168979, \quad (42)$$

$$c_{10}^{13\text{TeV}}(\kappa_{3t}) = -0.865038 - 3.92841 \kappa_{3t}, \quad (43)$$

$$c_{11}^{13\text{TeV}}(\kappa_{3t}) = -0.167158, \quad (44)$$

$$c_{20}^{13\text{TeV}}(\kappa_{3t}) = 0.872294 + 0.900894 \kappa_{3t}, \quad (45)$$

$$c_{21}^{13\text{ TeV}}(\kappa_{3t}) = 0.0487935, \quad (46)$$

$$c_{30}^{13\text{ TeV}}(\kappa_{3t}) = -0.259375, \quad (47)$$

$$c_{40}^{13\text{ TeV}}(\kappa_{3t}) = 0.0395354. \quad (48)$$

For $\sqrt{s} = 13.6\text{ TeV}$, they are

$$c_{00}^{13.6\text{ TeV}}(\kappa_{3t}) = 1 - 2.87872 \kappa_{3t} + 16.21306 \kappa_{3t}^2, \quad (49)$$

$$c_{01}^{13.6\text{ TeV}}(\kappa_{3t}) = -0.0909154 + 0.800038 \kappa_{3t}, \quad (50)$$

$$c_{02}^{13.6\text{ TeV}}(\kappa_{3t}) = 0.0168638, \quad (51)$$

$$c_{10}^{13.6\text{ TeV}}(\kappa_{3t}) = -0.866564 - 3.95165 \kappa_{3t}, \quad (52)$$

$$c_{11}^{13.6\text{ TeV}}(\kappa_{3t}) = -0.166501, \quad (53)$$

$$c_{20}^{13.6\text{ TeV}}(\kappa_{3t}) = 0.871502 + 0.897150 \kappa_{3t}, \quad (54)$$

$$c_{21}^{13.6\text{ TeV}}(\kappa_{3t}) = 0.0485042, \quad (55)$$

$$c_{30}^{13.6\text{ TeV}}(\kappa_{3t}) = -0.257216, \quad (56)$$

$$c_{40}^{13.6\text{ TeV}}(\kappa_{3t}) = 0.0390715. \quad (57)$$

For $\sqrt{s} = 14\text{ TeV}$, they are

$$c_{00}^{14\text{ TeV}}(\kappa_{3t}) = 1 - 2.95187 \kappa_{3t} + 16.47338 \kappa_{3t}^2, \quad (58)$$

$$c_{01}^{14\text{ TeV}}(\kappa_{3t}) = -0.0921387 + 0.803365 \kappa_{3t}, \quad (59)$$

$$c_{02}^{14\text{ TeV}}(\kappa_{3t}) = 0.0168553, \quad (60)$$

$$c_{10}^{14\text{ TeV}}(\kappa_{3t}) = -0.853992 - 3.93891 \kappa_{3t}, \quad (61)$$

$$c_{11}^{14\text{ TeV}}(\kappa_{3t}) = -0.166145, \quad (62)$$

$$c_{20}^{14\text{ TeV}}(\kappa_{3t}) = 0.863440 + 0.899872 \kappa_{3t}, \quad (63)$$

$$c_{21}^{14\text{ TeV}}(\kappa_{3t}) = 0.0482948, \quad (64)$$

$$c_{30}^{14\text{ TeV}}(\kappa_{3t}) = -0.255934, \quad (65)$$

$$c_{40}^{14\text{ TeV}}(\kappa_{3t}) = 0.0390078. \quad (66)$$

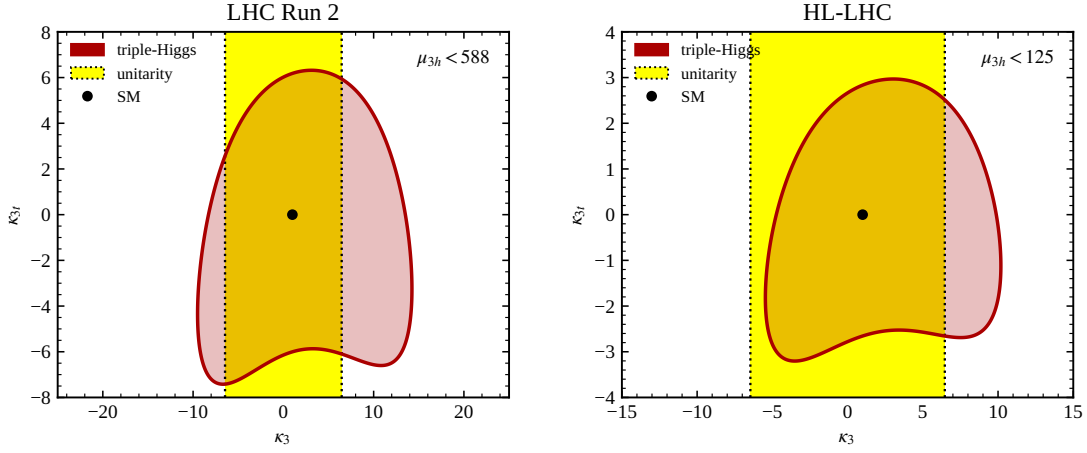


Figure 5: Constraints in the κ_3 - κ_{3t} plane at LHC Run 2 (left) and the HL-LHC (right), obtained from Eq. (38) using $\mu_{3h} < 588$ and $\mu_{3h} < 125$, respectively. The red regions are allowed by the inclusive triple-Higgs-boson rate, the black points denote the SM, and the yellow vertical bands show the projection of the tree-level $hh \rightarrow hh$ perturbative-unitarity constraint. We fix $\kappa_4 = \kappa_t = 1$ and $\kappa_{2t} = 0$. The yellow band constrains κ_3 only; no independent unitarity condition on κ_{3t} is imposed.

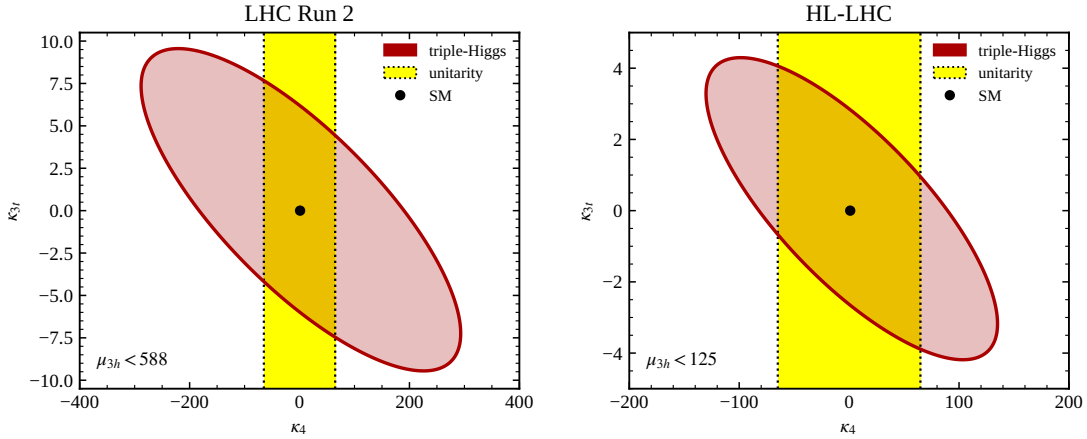


Figure 6: As in Figure 5, but in the κ_4 - κ_{3t} plane with $\kappa_3 = \kappa_t = 1$ and $\kappa_{2t} = 0$. The yellow band is the projection of the $hh \rightarrow hh$ perturbative-unitarity condition onto κ_4 .

Figures 5 and 6 are the κ_{3t} analogues of Figures 3 and 4. Fixing all other couplings to their SM values yields

$$-6.00 < \kappa_{3t} < 6.18 \quad (\text{LHC Run 2}), \quad -2.66 < \kappa_{3t} < 2.83 \quad (\text{HL-LHC}). \quad (67)$$

The quadratic form also exposes an inclusive-rate degeneracy. At $\kappa_3 = \kappa_4 = 1$, the two solutions of $c_{00}^{\sqrt{s}}(\kappa_{3t}) = 1$ are $\kappa_{3t} = 0$ and an energy-dependent second solution near $\kappa_{3t} = 0.179$. Differential information is therefore essential for distinguishing these two solutions.

Modifications of the Higgs boson trilinear and quartic self-couplings affect both the inclusive $gg \rightarrow 3h$ cross section and the kinematic properties of the final state. Figure 7 compares normalised distributions for the SM with two BSM benchmarks. Both benchmarks give $\mu_{3h} \simeq 65$, and their inclusive cross sections differ by less than 5%, while their values of κ_3 and κ_4 satisfy the perturbative-unitarity constraints from tree-level $hh \rightarrow hh$ scattering. We

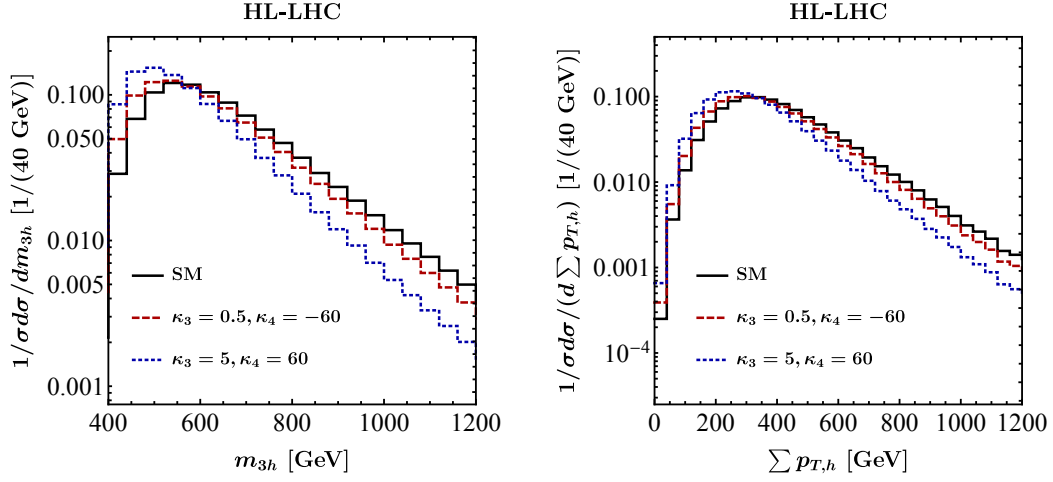


Figure 7: Normalised kinematic distributions for $g g \rightarrow 3h$ production at the HL-LHC in the SM (solid black line) and in two benchmark scenarios defined by κ_3 and κ_4 (red dashed and blue dotted lines). The left and right panels show the invariant mass of the **three-Higgs-boson** system, m_{3h} , and the scalar sum of the transverse momenta of the Higgs bosons, $\sum p_{T,h}$, respectively. All results are obtained for $\kappa_t = 1$ and $\kappa_{2t} = \kappa_{3t} = 0$.

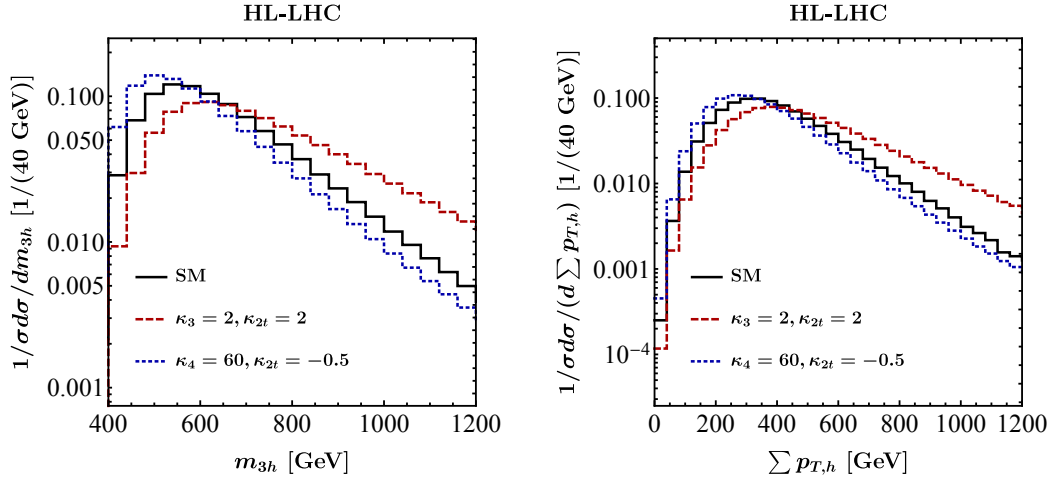


Figure 8: As in Figure 7, but for two **further benchmark** scenarios in the **κ -framework**, as indicated in the panels. The remaining parameters are fixed to their SM values, including $\kappa_{3t} = 0$.

set $\kappa_t = 1$ and $\kappa_{2t} = \kappa_{3t} = 0$. Variations of κ_t predominantly rescale the unnormalised spectra and therefore have little effect on their normalised shapes [41]. Nevertheless, the two benchmarks produce deviations of up to about 50% in the peak regions and high-energy tails of both distributions. Differential information can therefore strengthen future HL-LHC constraints beyond the projected interval in Eq. (7), which is obtained from the inclusive rate alone [21].

Figure 8 presents further normalised distributions for the SM and two BSM benchmarks in the κ -framework. Both benchmarks give $\mu_{3h} \simeq 75$, and their inclusive cross sections differ by less than 5%. The non-SM values of κ_3 or κ_4 satisfy the relevant perturbative-unitarity constraints, while the remaining parameters are fixed to their SM values, including $\kappa_{3t} = 0$.

As in Figure 7, sizeable **shape distortions remain** despite the near-degeneracy of the inclusive rates. This **illustrates how differential information can help to resolve parameter degeneracies** in future $gg \rightarrow 3h$ searches at the HL-LHC.

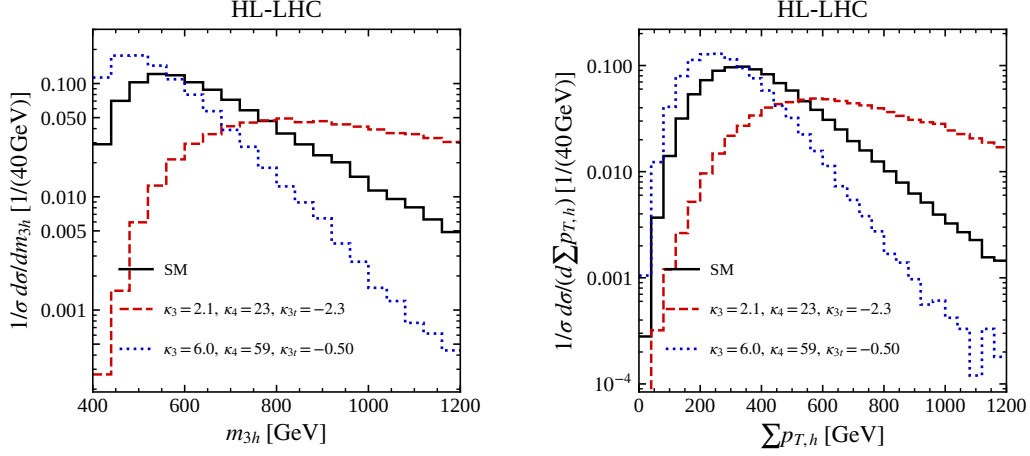


Figure 9: Normalised m_{3h} and $\sum p_{T,h}$ distributions at the HL-LHC for the SM and the two $\mu_{3h} \simeq 65$ benchmarks $(\kappa_3, \kappa_4, \kappa_{3t}) = (2.1, 23, -2.3)$ and $(6.0, 59, -0.50)$. The distributions use the same 40 GeV binning and plotting ranges as Figures 7 and 8. We fix $\kappa_t = 1$ and $\kappa_{2t} = 0$.

The two mixed-coupling benchmarks in Figure 9 provide a particularly clear example of enhanced but nearly degenerate inclusive rates. Their 100 000-event samples give $\mu_{3h} = 64.95$ and 64.80, respectively, so their inclusive cross sections differ by only 0.24%. Nevertheless, their normalised distributions differ substantially from both the SM and one another. The $(2.1, 23, -2.3)$ benchmark shifts weight towards larger invariant masses and transverse momenta, whereas the $(6.0, 59, -0.50)$ benchmark enhances the near-threshold and low- p_T regions. The samples use the same PDFs and scale choice as the inclusive-rate parametrisation.

3 Conclusions

Unlike the Higgs **boson** couplings to EW gauge bosons and third-generation fermions, which are already being measured with increasing precision, the Higgs **boson** self-interactions remain largely unconstrained by LHC Run 2 data. This leaves considerable room for BSM scenarios that modify the shape of the Higgs potential. Indeed, both theoretical considerations and explicit ultraviolet-complete constructions demonstrate that deviations of the Higgs **boson** self-couplings exceeding 200% can be consistent with existing **single-Higgs-boson** measurements without requiring fine-tuned parameter choices [4]. Measurements of **multi-Higgs-boson** production at the HL-LHC and future colliders therefore provide a unique opportunity to probe largely unexplored regions of parameter space and **offer** significant discovery potential, even in scenarios where **single-Higgs-boson** observables remain close to their SM predictions.

In this note, we have derived compact ‘**pocket formulae**’ for **triple-Higgs-boson** production within the κ -framework, enabling constraints on the Higgs **boson** trilinear and quartic self-couplings to be obtained from current and future measurements of the $gg \rightarrow 3h$ signal strength. The **parametrisation in Eq. (9)** retains the full dependence on $\kappa_3, \kappa_4, \kappa_t$, and κ_{2t} at LO in perturbation theory for $\kappa_{3t} = 0$. We have supplemented it with Eq. (38), which includes

the complete LO dependence on κ_{3t} for $\kappa_t = 1$ and $\kappa_{2t} = 0$ at 13, 13.6, and 14 TeV. Since the derivation relies on flexible matrix-element implementations, further extensions to incorporate additional coupling modifications — such as those arising at next-to-leading order in Higgs effective field theory [54] — are comparatively straightforward.

Using these parametrisations, we have studied how current LHC data and HL-LHC projections constrain κ_3 and κ_4 . We find that the presently allowed uncertainty in κ_t can have a substantial impact on the constraints in the κ_3 – κ_4 plane. This dependence is significantly reduced at the HL-LHC, where the expected percent-level precision of the top-quark Yukawa coupling measurement will strongly restrict the corresponding parameter space. For κ_{2t} , which parametrises the interaction between two top quarks and two Higgs bosons, we find that triple-Higgs-boson production is unlikely to provide stronger bounds than double-Higgs-boson production in a single-parameter analysis. Nevertheless, it can yield complementary information in global multi-parameter fits, especially once κ_4 is included, as this coupling remains only weakly constrained by double-Higgs-boson measurements [21–24]. In a single-parameter interpretation, the current and projected upper limits on the triple-Higgs-boson rate give $-6.00 < \kappa_{3t} < 6.18$ and $-2.66 < \kappa_{3t} < 2.83$, respectively. The energy-dependent nonzero solution near $\kappa_{3t} = 0.179$ is rate-degenerate with the SM. We further demonstrate that differential information lifts inclusive-rate degeneracies and identifies mixed $(\kappa_3, \kappa_4, \kappa_{3t})$ benchmarks with $\mu_{3h} \simeq 65$ but substantially different kinematic distributions from the SM and from one another.

The projected HL-LHC sensitivity to κ_4 in a single-parameter analysis remains modest, with constraints reaching only $|\kappa_4| \lesssim 80$, comparable to those derived from perturbative-unitarity arguments [34, 51, 52]. An improvement by approximately one order of magnitude in the determination of the Higgs boson quartic self-coupling could be achieved at the proposed hadron-collider stage of the Future Circular Collider [22, 28–30, 32, 33, 35–38, 40]. Reaching a precision at the level of 50%, however, would likely require the capabilities of a future muon collider [56].

Acknowledgements

UH acknowledges Konstantin Schmid for helpful discussions. This manuscript has been authored by Fermi Forward Discovery Group, LLC under Contract No. 89243024CSC000002 with the U.S. Department of Energy, Office of Science, Office of High Energy Physics. RKE acknowledges receipt of a Leverhulme Emeritus Fellowship from the Leverhulme Trust.

References

- [1] G. Aad *et al.*, *Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC*, Phys. Lett. B **716**, 1 (2012), doi:[10.1016/j.physletb.2012.08.020](https://doi.org/10.1016/j.physletb.2012.08.020), [1207.7214](https://arxiv.org/abs/1207.7214).
- [2] S. Chatrchyan *et al.*, *Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC*, Phys. Lett. B **716**, 30 (2012), doi:[10.1016/j.physletb.2012.08.021](https://doi.org/10.1016/j.physletb.2012.08.021), [1207.7235](https://arxiv.org/abs/1207.7235).
- [3] J. Alison *et al.*, *Higgs boson potential at colliders: Status and perspectives*, Rev. Phys. **5**, 100045 (2020), doi:[10.1016/j.revip.2020.100045](https://doi.org/10.1016/j.revip.2020.100045), [1910.00012](https://arxiv.org/abs/1910.00012).

- [4] G. Durieux, M. McCullough and E. Salvioni, *Charting the Higgs self-coupling boundaries*, JHEP **12**, 148 (2022), doi:[10.1007/JHEP12\(2022\)148](https://doi.org/10.1007/JHEP12(2022)148), [Erratum: JHEP **02**, 165 (2023)], [2209.00666](https://arxiv.org/abs/2209.00666).
- [5] M. Gorbahn and U. Haisch, *Indirect probes of the trilinear Higgs coupling: $gg \rightarrow h$ and $h \rightarrow \gamma\gamma$* , JHEP **10**, 094 (2016), doi:[10.1007/JHEP10\(2016\)094](https://doi.org/10.1007/JHEP10(2016)094), [1607.03773](https://arxiv.org/abs/1607.03773).
- [6] G. Degrandi, P. P. Giardino, F. Maltoni and D. Pagani, *Probing the Higgs self coupling via single Higgs production at the LHC*, JHEP **12**, 080 (2016), doi:[10.1007/JHEP12\(2016\)080](https://doi.org/10.1007/JHEP12(2016)080), [1607.04251](https://arxiv.org/abs/1607.04251).
- [7] W. Bizon, M. Gorbahn, U. Haisch and G. Zanderighi, *Constraints on the trilinear Higgs coupling from vector boson fusion and associated Higgs production at the LHC*, JHEP **07**, 083 (2017), doi:[10.1007/JHEP07\(2017\)083](https://doi.org/10.1007/JHEP07(2017)083), [1610.05771](https://arxiv.org/abs/1610.05771).
- [8] F. Maltoni, D. Pagani, A. Shivaji and X. Zhao, *Trilinear Higgs coupling determination via single-Higgs differential measurements at the LHC*, Eur. Phys. J. C **77**(12), 887 (2017), doi:[10.1140/epjc/s10052-017-5410-8](https://doi.org/10.1140/epjc/s10052-017-5410-8), [1709.08649](https://arxiv.org/abs/1709.08649).
- [9] M. Gorbahn and U. Haisch, *Two-loop amplitudes for Higgs plus jet production involving a modified trilinear Higgs coupling*, JHEP **04**, 062 (2019), doi:[10.1007/JHEP04\(2019\)062](https://doi.org/10.1007/JHEP04(2019)062), [1902.05480](https://arxiv.org/abs/1902.05480).
- [10] U. Haisch and G. Koole, *Off-shell Higgs production at the LHC as a probe of the trilinear Higgs coupling*, JHEP **02**, 030 (2022), doi:[10.1007/JHEP02\(2022\)030](https://doi.org/10.1007/JHEP02(2022)030), [2111.12589](https://arxiv.org/abs/2111.12589).
- [11] J. Gao, X.-M. Shen, G. Wang, L. L. Yang and B. Zhou, *Probing the Higgs boson trilinear self-coupling through Higgs boson+jet production*, Phys. Rev. D **107**(11), 115017 (2023), doi:[10.1103/PhysRevD.107.115017](https://doi.org/10.1103/PhysRevD.107.115017), [2302.04160](https://arxiv.org/abs/2302.04160).
- [12] U. Haisch and M. Niggetiedt, *Exact two-loop amplitudes for Higgs plus jet production with a cubic Higgs self-coupling*, JHEP **10**, 236 (2024), doi:[10.1007/JHEP10\(2024\)236](https://doi.org/10.1007/JHEP10(2024)236), [2408.13186](https://arxiv.org/abs/2408.13186).
- [13] A. Ghosh, M. Griesse, U. Haisch and T. H. Park, *Neural simulation-based inference of the Higgs trilinear self-coupling via off-shell Higgs production*, Eur. Phys. J. C **86**(4), 415 (2026), doi:[10.1140/epjc/s10052-026-15605-3](https://doi.org/10.1140/epjc/s10052-026-15605-3), [2507.02032](https://arxiv.org/abs/2507.02032).
- [14] A. David, A. Denner, M. Duehrssen, M. Grazzini, C. Grojean, G. Passarino, M. Schumacher, M. Spira, G. Weiglein and M. Zanetti, *LHC HXSWG interim recommendations to explore the coupling structure of a Higgs-like particle* (2012), [1209.0040](https://arxiv.org/abs/1209.0040).
- [15] J. R. Andersen et al., *Handbook of LHC Higgs Cross Sections: 3. Higgs Properties* (2013), doi:[10.5170/CERN-2013-004](https://doi.org/10.5170/CERN-2013-004), [1307.1347](https://arxiv.org/abs/1307.1347).
- [16] D. de Florian et al., *Handbook of LHC Higgs Cross Sections: 4. Deciphering the Nature of the Higgs Sector*, CERN Yellow Rep. Monogr. **2**, 1 (2017), doi:[10.23731/CYRM-2017-002](https://doi.org/10.23731/CYRM-2017-002), [1610.07922](https://arxiv.org/abs/1610.07922).
- [17] G. Aad et al., *Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV* (2026), [2602.23991](https://arxiv.org/abs/2602.23991).
- [18] D. de Florian, I. Fabre and J. Mazzitelli, *Triple Higgs production at hadron colliders at NNLO in QCD*, JHEP **03**, 155 (2020), doi:[10.1007/JHEP03\(2020\)155](https://doi.org/10.1007/JHEP03(2020)155), [1912.02760](https://arxiv.org/abs/1912.02760).

- [19] G. Aad *et al.*, *Search for triple Higgs boson production in the $6b$ final state using pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, Phys. Rev. D **111**(3), 032006 (2025), doi:[10.1103/PhysRevD.111.032006](https://doi.org/10.1103/PhysRevD.111.032006), [2411.02040](https://arxiv.org/abs/2411.02040).
- [20] *Search for nonresonant triple Higgs boson production in the six b -quark final state in proton-proton collisions at 13 TeV*, CERN, Geneva (2025), <https://cds.cern.ch/record/2945361>.
- [21] J.-L. Ding *et al.*, *Constraining the Higgs potential using multi-Higgs production* (2026), doi:[10.21468/SciPostPhysCommRep.24.2601.13248](https://doi.org/10.21468/SciPostPhysCommRep.24.2601.13248).
- [22] W. Bizoń, U. Haisch and L. Rottoli, *Constraints on the quartic Higgs self-coupling from double-Higgs production at future hadron colliders*, JHEP **10**, 267 (2019), doi:[10.1007/JHEP10\(2019\)267](https://doi.org/10.1007/JHEP10(2019)267), [1810.04665](https://arxiv.org/abs/1810.04665).
- [23] S. Borowka, C. Duhr, F. Maltoni, D. Pagani, A. Shivaji and X. Zhao, *Probing the scalar potential via double Higgs boson production at hadron colliders*, JHEP **04**, 016 (2019), doi:[10.1007/JHEP04\(2019\)016](https://doi.org/10.1007/JHEP04(2019)016), [1811.12366](https://arxiv.org/abs/1811.12366).
- [24] H. T. Li, Z.-G. Si, J. Wang, X. Zhang and D. Zhao, *Improved constraints on Higgs boson self-couplings with quartic and cubic power dependencies of the cross section*, Chin. Phys. C **49**(2), 023107 (2025), doi:[10.1088/1674-1137/ad9d1d](https://doi.org/10.1088/1674-1137/ad9d1d), [2407.14716](https://arxiv.org/abs/2407.14716).
- [25] T. Plehn and M. Rauch, *The quartic Higgs coupling at hadron colliders*, Phys. Rev. D **72**, 053008 (2005), doi:[10.1103/PhysRevD.72.053008](https://doi.org/10.1103/PhysRevD.72.053008), [hep-ph/0507321](https://arxiv.org/abs/hep-ph/0507321).
- [26] T. Binoth, S. Karg, N. Kauer and R. Ruckl, *Multi-Higgs boson production in the Standard Model and beyond*, Phys. Rev. D **74**, 113008 (2006), doi:[10.1103/PhysRevD.74.113008](https://doi.org/10.1103/PhysRevD.74.113008), [hep-ph/0608057](https://arxiv.org/abs/hep-ph/0608057).
- [27] F. Maltoni, E. Vryonidou and M. Zaro, *Top-quark mass effects in double and triple Higgs production in gluon-gluon fusion at NLO*, JHEP **11**, 079 (2014), doi:[10.1007/JHEP11\(2014\)079](https://doi.org/10.1007/JHEP11(2014)079), [1408.6542](https://arxiv.org/abs/1408.6542).
- [28] A. Papaefstathiou and K. Sakurai, *Triple Higgs boson production at a 100 TeV proton-proton collider*, JHEP **02**, 006 (2016), doi:[10.1007/JHEP02\(2016\)006](https://doi.org/10.1007/JHEP02(2016)006), [1508.06524](https://arxiv.org/abs/1508.06524).
- [29] C.-Y. Chen, Q.-S. Yan, X. Zhao, Y.-M. Zhong and Z. Zhao, *Probing triple-Higgs productions via $4b2\gamma$ decay channel at a 100 TeV hadron collider*, Phys. Rev. D **93**(1), 013007 (2016), doi:[10.1103/PhysRevD.93.013007](https://doi.org/10.1103/PhysRevD.93.013007), [1510.04013](https://arxiv.org/abs/1510.04013).
- [30] B. Fuks, J. H. Kim and S. J. Lee, *Probing Higgs self-interactions in proton-proton collisions at a center-of-mass energy of 100 TeV*, Phys. Rev. D **93**(3), 035026 (2016), doi:[10.1103/PhysRevD.93.035026](https://doi.org/10.1103/PhysRevD.93.035026), [1510.07697](https://arxiv.org/abs/1510.07697).
- [31] D. de Florian and J. Mazzitelli, *Two-loop corrections to the triple Higgs boson production cross section*, JHEP **02**, 107 (2017), doi:[10.1007/JHEP02\(2017\)107](https://doi.org/10.1007/JHEP02(2017)107), [1610.05012](https://arxiv.org/abs/1610.05012).
- [32] W. Kilian, S. Sun, Q.-S. Yan, X. Zhao and Z. Zhao, *New Physics in multi-Higgs boson final states*, JHEP **06**, 145 (2017), doi:[10.1007/JHEP06\(2017\)145](https://doi.org/10.1007/JHEP06(2017)145), [1702.03554](https://arxiv.org/abs/1702.03554).
- [33] B. Fuks, J. H. Kim and S. J. Lee, *Scrutinizing the Higgs quartic coupling at a future 100 TeV proton-proton collider with taus and b -jets*, Phys. Lett. B **771**, 354 (2017), doi:[10.1016/j.physletb.2017.05.075](https://doi.org/10.1016/j.physletb.2017.05.075), [1704.04298](https://arxiv.org/abs/1704.04298).
- [34] P. Stylianou and G. Weiglein, *Constraints on the trilinear and quartic Higgs couplings from triple Higgs production at the LHC and beyond*, Eur. Phys. J. C **84**(4), 366 (2024), doi:[10.1140/epjc/s10052-024-12722-9](https://doi.org/10.1140/epjc/s10052-024-12722-9), [2312.04646](https://arxiv.org/abs/2312.04646).

- [35] A. Papaefstathiou and G. Tetlalmatzi-Xolocotzi, *Multi-Higgs boson production with anomalous interactions at current and future proton colliders*, JHEP **06**, 124 (2024), doi:[10.1007/JHEP06\(2024\)124](https://doi.org/10.1007/JHEP06(2024)124), [2312.13562](https://arxiv.org/abs/2312.13562).
- [36] H. Abouabid *et al.*, *HHH whitepaper*, Eur. Phys. J. C **84**, 1183 (2024), doi:[10.1140/epjc/s10052-024-13376-3](https://doi.org/10.1140/epjc/s10052-024-13376-3), [2407.03015](https://arxiv.org/abs/2407.03015).
- [37] Z. Dong, X. Sun, B. Guo, L. Zhang, Z. Li, J. Wang, Z. Li, Y. Ban and Y. Mao, *Probing triple Higgs production via $4\tau 2b$ decay channel at a 100 TeV hadron collider*, JHEP **08**, 040 (2025), doi:[10.1007/JHEP08\(2025\)040](https://doi.org/10.1007/JHEP08(2025)040), [2504.04037](https://arxiv.org/abs/2504.04037).
- [38] U. Haisch, A. Sankar and G. Zanderighi, *A new probe of the quartic Higgs self-coupling* (2025), [2505.20463](https://arxiv.org/abs/2505.20463).
- [39] J. M. Campbell, G. De Laurentis and R. K. Ellis, *An analytic result for the $0 \rightarrow ggHHH$ amplitude*, JHEP **01**, 157 (2026), doi:[10.1007/JHEP01\(2026\)157](https://doi.org/10.1007/JHEP01(2026)157), [2507.19313](https://arxiv.org/abs/2507.19313).
- [40] B. Fuks, A. Papaefstathiou and G. Tetlalmatzi-Xolocotzi, *Extracting Higgs self-coupling constraints through triple Higgs boson production at future hadron colliders*, Eur. Phys. J. C **85**(11), 1309 (2025), doi:[10.1140/epjc/s10052-025-15051-7](https://doi.org/10.1140/epjc/s10052-025-15051-7), [2509.16364](https://arxiv.org/abs/2509.16364).
- [41] L. Panizzi, *Role of the top Yukawa coupling in triple Higgs production at the LHC*, Phys. Rev. D **113**(5), L051706 (2026), doi:[10.1103/5ssz-3nh7](https://doi.org/10.1103/5ssz-3nh7), [2512.03003](https://arxiv.org/abs/2512.03003).
- [42] J. M. Campbell and R. K. Ellis, *An Update on vector boson pair production at hadron colliders*, Phys. Rev. D **60**, 113006 (1999), doi:[10.1103/PhysRevD.60.113006](https://doi.org/10.1103/PhysRevD.60.113006), [hep-ph/9905386](https://arxiv.org/abs/hep-ph/9905386).
- [43] J. M. Campbell, R. K. Ellis and C. Williams, *Vector Boson Pair Production at the LHC*, JHEP **07**, 018 (2011), doi:[10.1007/JHEP07\(2011\)018](https://doi.org/10.1007/JHEP07(2011)018), [1105.0020](https://arxiv.org/abs/1105.0020).
- [44] J. M. Campbell, R. K. Ellis and W. T. Giele, *A Multi-Threaded Version of MCFM*, Eur. Phys. J. C **75**(6), 246 (2015), doi:[10.1140/epjc/s10052-015-3461-2](https://doi.org/10.1140/epjc/s10052-015-3461-2), [1503.06182](https://arxiv.org/abs/1503.06182).
- [45] J. Campbell and T. Neumann, *Precision Phenomenology with MCFM*, JHEP **12**, 034 (2019), doi:[10.1007/JHEP12\(2019\)034](https://doi.org/10.1007/JHEP12(2019)034), [1909.09117](https://arxiv.org/abs/1909.09117).
- [46] R. D. Ball *et al.*, *The PDF4LHC21 combination of global PDF fits for the LHC Run III*, J. Phys. G **49**(8), 080501 (2022), doi:[10.1088/1361-6471/ac7216](https://doi.org/10.1088/1361-6471/ac7216), [2203.05506](https://arxiv.org/abs/2203.05506).
- [47] *HL-LHC prospects for the measurement of triple-Higgs production in the $6b$ final state at the ATLAS experiment*, CERN, Geneva, (2025), <https://cds.cern.ch/record/2924772>.
- [48] G. Aad *et al.*, *A detailed map of Higgs boson interactions by the ATLAS experiment ten years after the discovery*, Nature **607**(7917), 52 (2022), doi:[10.1038/s41586-022-04893-w](https://doi.org/10.1038/s41586-022-04893-w), [Erratum: Nature **612**, E24 (2022)], [2207.00092](https://arxiv.org/abs/2207.00092).
- [49] A. Tumasyan *et al.*, *A portrait of the Higgs boson by the CMS experiment ten years after the discovery*, Nature **607**(7917), 60 (2022), doi:[10.1038/s41586-022-04892-x](https://doi.org/10.1038/s41586-022-04892-x), [Erratum: Nature **623**, (2023)], [2207.00043](https://arxiv.org/abs/2207.00043).
- [50] *Highlights of the HL-LHC physics projections by ATLAS and CMS*, CERN, Geneva, (2025), <https://cds.cern.ch/record/2928907>.
- [51] T. Liu, K.-F. Lyu, J. Ren and H. X. Zhu, *Probing the quartic Higgs boson self-interaction*, Phys. Rev. D **98**(9), 093004 (2018), doi:[10.1103/PhysRevD.98.093004](https://doi.org/10.1103/PhysRevD.98.093004), [1803.04359](https://arxiv.org/abs/1803.04359).

- [52] L. Di Luzio, R. Gröber and M. Spannowsky, *Maxi-sizing the trilinear Higgs self-coupling: how large could it be?*, Eur. Phys. J. C **77**(11), 788 (2017), doi:[10.1140/epjc/s10052-017-5361-0](https://doi.org/10.1140/epjc/s10052-017-5361-0), [1704.02311](https://arxiv.org/abs/1704.02311).
- [53] G. Buchalla, M. Capozzi, A. Celis, G. Heinrich and L. Scyboz, *Higgs boson pair production in non-linear Effective Field Theory with full m_t -dependence at NLO QCD*, JHEP **09**, 057 (2018), doi:[10.1007/JHEP09\(2018\)057](https://doi.org/10.1007/JHEP09(2018)057), [Erratum: JHEP **06**, 094 (2025)], [1806.05162](https://arxiv.org/abs/1806.05162).
- [54] I. Brivio, R. Gröber and K. Schmid, *Higgs pair production in gluon fusion to higher orders in Higgs Effective Field Theory*, JHEP **05**, 031 (2026), doi:[10.1007/JHEP05\(2026\)031](https://doi.org/10.1007/JHEP05(2026)031), [2511.23411](https://arxiv.org/abs/2511.23411).
- [55] J. Alwall *et al.*, *The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations*, JHEP **07**, 079 (2014), doi:[10.1007/JHEP07\(2014\)079](https://doi.org/10.1007/JHEP07(2014)079), [1405.0301](https://arxiv.org/abs/1405.0301).
- [56] M. Chiesa, F. Maltoni, L. Mantani, B. Mele, F. Piccinini and X. Zhao, *Measuring the quartic Higgs self-coupling at a multi-TeV muon collider*, JHEP **09**, 098 (2020), doi:[10.1007/JHEP09\(2020\)098](https://doi.org/10.1007/JHEP09(2020)098), [2003.13628](https://arxiv.org/abs/2003.13628).